A LENGTH-14 NEGATIVE ANSWER TO ALDERSON'S $q = 4$ OPTIMALITY QUESTION

Abstract. Alderson constructed an extendable, additively maximal additive $(112, 2, 104)_{16/4}$-code and asked whether length 112 is optimal for $q = 4$. A classical spread of $\mathrm{PG}(3, 4) \setminus \mathrm{PG}(3, 2)$ yields a faithful additive MDS $(14, 2, 13)_{16/4}$-code that is extendable as a 16-ary code but admits no $\mathbb{F}_4$-additive extension. Hence the answer is negative and the minimum length in the $q = 4$ clause of Alderson's Problem 9.2 is at most 14. Problem 9.2 itself notes that any sub-multiset of its 112 external lines whose maximal-fold points still meet every line yields a shorter example; that condition governs additive maximality alone, and what is supplied here is a sub-multiset for which extendability, the other requirement, also holds. We do not determine the global minimum.

1. Result

Alderson's Theorem 6.6 gives, for $t = 2$, an extendable, additively maximal additive $(112, 2, 104)_{16/4}$-code, whose 112 coordinates are indexed by the lines of $\Pi = \mathrm{PG}(3, 4)$ external to a scattered $\mathbb{F}_2$-linear set. Problem 9.2 of [1] then reads, in full:

> Extend Theorem 6.6 to non-square, non-prime q. Further, determine the minimum length of an extendable, additively maximal additive $(n, 2, d)_{q^2/q}$-code for square q: is $n = 112$ optimal for $q = 4$? (Any sub-multiset of the external lines whose set of maximal-fold points still meets every line of Π yields a shorter example.)

Theorem 1 below answers the final clause negatively, at length 14; the first clause and the minimum for general square q are untouched. Remark 3 records what the closing parenthetical already supplies and what it does not.

The geometric input is classical: van Dam [3] classified the spreads of $\mathrm{PG}(3, 4) \setminus \mathrm{PG}(3, 2)$ into three orbits under the collineation group $\mathrm{P\Gamma L}(4, 4)$, and Mellinger [4] later obtained the corresponding three mixed partitions consisting of one Baer subgeometry and fourteen lines.

Theorem 1. *There exists a faithful additive MDS $(14, 2, 13)_{16/4}$-code C that is extendable as a 16-ary code but is additively maximal. Its Hamming weight enumerator is*

$$W_C(z) = 1 + 210z^{13} + 45z^{14}.$$

2020 *Mathematics Subject Classification.* 94B05, 51E20.

Key words and phrases. additive code, code extension, maximal code, partial spread, Baer subgeometry.

Proof. Let $\mathbb{F}_2 \subset \mathbb{F}_4 \subset \mathbb{F}_{16}$, choose $\omega \in \mathbb{F}_4$ with $\omega^2 + \omega + 1 = 0$, put $E = \mathbb{F}_4^4$, and let $U = \mathbb{F}_2^4 \le E$. In $\Pi = \mathrm{PG}(E) = \mathrm{PG}(3,4)$, define the standard Baer subgeometry

$$B = \{\langle u\rangle_{\mathbb{F}_4} : 0 \ne u \in U\} \cong \mathrm{PG}(3,2).$$

Thus $|B| = 15$. The $\mathbb{F}_4$-line through two distinct points of B meets B in the three points of the corresponding $\mathbb{F}_2$-subline. In particular, B contains no $\mathbb{F}_4$-line.

We first give an explicit representative of the classical complement spreads classified in [3, 4]. Let

$$T = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \in \mathrm{GL}(4,2) \le \mathrm{GL}(4,4).$$

The upper-left block has minimal polynomial $X^3 + X + 1$, so T has order 7. Put

$$u = (1,0,0,\omega)^{\mathsf{T}}, \quad v = (0,1,\omega^2,0)^{\mathsf{T}}, \quad L = \langle u, v\rangle_{\mathbb{F}_4},$$
$$u' = (1,0,0,\omega^2)^{\mathsf{T}}, \quad v' = (0,1,\omega,0)^{\mathsf{T}}, \quad L' = \langle u', v'\rangle_{\mathbb{F}_4}.$$

The five points of $\mathrm{PG}(L)$ are represented by $u, v, u+v, u+\omega v, u+\omega^2 v$. These representatives are normalized at their first nonzero coordinate, and each has a coordinate outside $\mathbb{F}_2$; hence none lies in B. The same holds for $\mathrm{PG}(L')$, its Frobenius conjugate. Since T has entries in $\mathbb{F}_2$, every line $\mathrm{PG}(T^j L)$ and $\mathrm{PG}(T^j L')$ is disjoint from B.

For vectors $x_1, \dots, x_4$, write $\det[x_1, x_2, x_3, x_4]$ for the determinant of the matrix with those columns. A direct calculation gives

$$\big(\det[u, v, T^k u, T^k v]\big)_{k=1}^{6} = (\omega, 1, \omega^2, \omega^2, 1, \omega),$$
$$\big(\det[u', v', T^k u', T^k v']\big)_{k=1}^{6} = (\omega^2, 1, \omega, \omega, 1, \omega^2),$$
$$\big(\det[u, v, T^k u', T^k v']\big)_{k=0}^{6} = (1, \omega, \omega, \omega^2, \omega, \omega^2, \omega^2).$$

All entries are nonzero. Since intersections depend only on relative powers of T, these three calculations show that the fourteen lines

$$\ell_{j+1} = \mathrm{PG}(T^j L), \qquad \ell_{j+8} = \mathrm{PG}(T^j L') \qquad (0 \le j \le 6)$$

are pairwise disjoint. They contain $14 \cdot 5 = 70 = 85 - 15$ points, all outside B, and therefore

$$\Pi = B \,\dot\cup\, \ell_1 \,\dot\cup \cdots \dot\cup\, \ell_{14}. \tag{1}$$

Write $\ell_i = \mathrm{PG}(V_i)$, where $V_i \le E$ has $\mathbb{F}_4$-dimension 2. Identifying $\mathbb{F}_{16}$ with a two-dimensional $\mathbb{F}_4$-space, choose an $\mathbb{F}_4$-linear epimorphism

$$\rho_i : E \longrightarrow \mathbb{F}_{16}, \qquad \ker \rho_i = V_i,$$

and define

$$c(a) = (\rho_1(a), \dots, \rho_{14}(a)), \qquad C = \{c(a) : a \in E\}.$$

Since the lines are disjoint,

$$\ker c = \bigcap_{i=1}^{14} V_i \subseteq V_1 \cap V_2 = \{0\}.$$

Hence c is injective and $|C| = 4^4 = 16^2$. Each coordinate map is surjective, so C is faithful.

For $a \neq 0$, the zero coordinates of $c(a)$ are exactly the indices i for which $\langle a \rangle_{\mathbb{F}_4} \in \ell_i$. By (1), $c(a)$ has weight 14 when $\langle a \rangle_{\mathbb{F}_4} \in B$ and weight 13 otherwise. Since $|\Pi| = 85$, there are $3|B| = 45$ nonzero words of weight 14 and $3(85-15) = 210$ of weight 13. This proves the stated weight enumerator and minimum distance. Since $13 = 14 - 2 + 1$, the code is MDS.

Viewed over $\mathbb{F}_2$, both E/U and $\mathbb{F}_{16}$ have dimension 4. Choose an $\mathbb{F}_2$-linear isomorphism $\tau : E/U \to \mathbb{F}_{16}$ and set, following the coset argument in the proof of [1, Theorem 6.6],

$$C^+ = \{(c(a), \tau(a+U)) : a \in E\} \subseteq \mathbb{F}_{16}^{15}.$$

For distinct $a, b \in E$, put $x = a - b$. If $x \in U$, then the new symbols agree and $\langle x \rangle_{\mathbb{F}_4} \in B$, so $d_H(c(a), c(b)) = 14$. If $x \notin U$, then the new symbols differ and $d_H(c(a), c(b)) \geq 13$. Thus $d(C^+) = 14$, with equality in the first case, and puncturing the last coordinate gives C. Hence C is extendable; in fact, this extension is $\mathbb{F}_2$-linear, the subfield phenomenon of [1, Remark 6.7]. So C, regarded not as an $\mathbb{F}_4$-linear but as an $\mathbb{F}_2$-linear subspace of $\mathbb{F}_{16}^{14}$ — that is, as an additive $(14, 2, 13)_{16/2}$-code in the general $(n, k, d)_{q^m/q}$ notation of [1], taken there with $q = 2$ and $m = 4$, so that the $\mathbb{F}_2$-dimension is $km = 8$, as it is here since $|C| = 2^8$ — does admit an additive extension: the additive maximality proved next is relative to the declared field of linearity $\mathbb{F}_4$.

Suppose that C admitted an $\mathbb{F}_4$-additive extension $\widetilde{C} \leq \mathbb{F}_{16}^{15}$ of minimum distance 14. Puncturing the new coordinate is an $\mathbb{F}_4$-linear bijection $\widetilde{C} \to C$. Therefore, for some $\mathbb{F}_4$-linear map $\rho : E \to \mathbb{F}_{16}$,

$$\widetilde{C} = \{(c(a), \rho(a)) : a \in E\}.$$

If $\langle a \rangle_{\mathbb{F}_4} \notin B$, then $\mathrm{wt}(c(a)) = 13$, so $\rho(a) \neq 0$. Consequently every nonzero vector in $\ker \rho$ has projective direction in B. But $\dim_{\mathbb{F}_4} \mathbb{F}_{16} = 2$, and hence

$$\dim_{\mathbb{F}_4} \ker \rho \geq 4 - 2 = 2.$$

The projectivization of a two-dimensional $\mathbb{F}_4$-subspace of $\ker \rho$ is an $\mathbb{F}_4$-line contained in B, a contradiction. Thus C has no $\mathbb{F}_4$-additive extension. $\square$

Corollary 2. *The minimum length asked for in the $q = 4$ clause of Alderson's Problem 9.2 is at most 14. In particular, 112 is not optimal.*

Proof. Theorem 1 has exactly the required extension and additive maximality properties. $\square$

Remark 3. Problem 9.2 already states that any sub-multiset of the 112 external lines whose set of maximal-fold points still meets every line of Π yields a shorter example, and the code of Theorem 1 is such a sub-multiset:

each of $\ell_1, \ldots, \ell_{14}$ is taken once, so by (1) the maximal fold is 1 and the maximal-fold set is $\Pi \setminus B$, which meets every line of Π because B contains no $\mathbb{F}_4$-line. For $t = 2$ Alderson's scattered $\mathbb{F}_2$-linear set is the projectivization of $\{(x, y, x^2, y^2) : x, y \in \mathbb{F}_4\}$, an $\mathbb{F}_2$-subspace of E whose $\mathbb{F}_2$-basis is also an $\mathbb{F}_4$-basis of E, hence the image of $U = \mathbb{F}_2^4$ under an element of $\mathrm{GL}(4, 4)$; it is therefore a Baer subgeometry $\mathrm{PG}(3, 2)$ of Π (cf. [1, Lemma 6.4]), projectively equivalent to the set B above. The lines of Π external to a Baer subgeometry number exactly 112: of the 357 lines of Π, 112 miss B, 210 meet it in one point and 35 in three. That count is Alderson's $n_2 = 112$, so $\ell_1, \ldots, \ell_{14}$ are fourteen of his own 112 external lines, each taken once; the maximal-fold set $\Pi \setminus B$ is his; and C^+ is his coset construction.

What that parenthetical supplies is one of the two requirements. Its condition, that the maximal-fold points meet every line of Π, is the criterion for additive maximality of [1, Corollary 3.6], in which the flats to be met are the $(km - m - 1)$-flats of Π and hence, for $k = m = 2$, its lines; it is silent on extendability, which Problem 9.2 also demands, so a sub-multiset meeting the condition need not answer the problem. The content of Theorem 1 is accordingly the choice of sub-multiset: a line–Baer partition of Π makes the fourteen chosen lines pairwise disjoint, so the code is MDS, and leaves the uncovered points equal to the Baer subgeometry B, and it is that hole set which makes the 16 cosets of U available as the fibres of a 16-ary extension of minimum distance 14.

Remark 4. The corollary answers only the final yes-or-no clause of Alderson's Problem 9.2, refuting the optimality of 112 for $q = 4$. It does not determine the global minimum or address the nonsquare, nonprime case, and no lower bound is proved here. The spread in (1) is classical; the contribution is its coding-theoretic consequence in Alderson's extension framework. In the projective-systems dictionary used in [1], a faithful additive MDS $(n, 2, n-1)_{q^2/q}$-code is a partial spread of n lines of $\mathrm{PG}(3, q)$, and additive maximality corresponds, by [1, Corollary 3.6] again, to that partial spread being maximal; the object here is thus a maximal partial spread of $\mathrm{PG}(3, 4)$ of size 14 whose holes form a Baer subgeometry. Additive MDS codes over $\mathbb{F}_{16}$ that are $\mathbb{F}_4$-linear are studied in [2], where the longest ones are classified; additive extendability in the sense of [1] is not considered there.

Exact verification

Everything needed to redo the computation is printed above: the relation $\omega^2 + \omega + 1 = 0$, the matrix T, the vectors u, v, u', v', and the partition (1). From these one forms the fourteen lines $\ell_{j+1} = \mathrm{PG}(T^j L)$ and $\ell_{j+8} = \mathrm{PG}(T^j L')$, checks the nineteen determinants displayed in the proof, tests all 91 pairs of lines for disjointness, builds c, C and C^+ as in the proof, recovers $W_C(z) = 1 + 210z^{13} + 45z^{14}$ and the minimum distances $d(C) = 13$ and $d(C^+) = 14$, and finally runs over the $\mathbb{F}_4$-linear maps $\rho : E \to \mathbb{F}_{16}$ to confirm that none of them extends C to minimum distance 14. A referee can therefore reproduce every number printed here from this paper alone.

The exhibited object and every count printed above were recomputed on that basis independently, on a separate machine, by a program written from the definitions of this note; all 66 of its checks agree with the paper. The enumerations are exhaustive rather than sampled: all 85 points and all 357 lines of Π, all 91 pairs of the fourteen lines, all 256 words of C and of C^+ with all 32,640 unordered pairs for each minimum distance, all 357 two-dimensional $\mathbb{F}_4$-subspaces of E, and, in the maximality step, all $4^8 = 65{,}536$ $\mathbb{F}_4$-linear maps $E \to \mathbb{F}_{16}$, the completeness of that last list being itself checked. The run repeats the construction for other choices of $\rho_1, \dots, \rho_{14}$ and of τ, reports failures on seven deliberately corrupted variants of the object, and passes a positive control in which the hole set does contain a line: replacing $\ell_1, \dots, \ell_{14}$ by fourteen of the seventeen lines of a spread of $\mathrm{PG}(3,4)$ leaves as holes the three remaining, pairwise disjoint lines, and the same exhaustive search then accepts exactly $3\,|\mathrm{GL}(2,4)| = 540$ of the 65,536 maps ρ, namely, for each hole line $\mathrm{PG}(V)$, the 180 maps with kernel V. So the negative answer for the Baer hole set is not an artifact of the search. The program and its transcript are retained in an auxiliary archive, available on request; nothing above depends on them, since every quantity quoted is either printed here or recomputable from the data just listed.

What the re-run does not do. It does not re-derive anything from [1]: Theorem 6.6, the length 112, and the definitions of extendable and of additively maximal are taken from that paper. It does not verify the classifications of [3, 4], for which the explicit spread above is a self-contained substitute. And faithfulness is tested only in the sense used here, that every coordinate projection is onto $\mathbb{F}_{16}$. No search for a lower bound or for the global minimum was attempted. The re-run and this paper agree; that is agreement, not a proof.

References

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